The sharp corner threshold for five-colour cubes

Jason Davies

November 2, 2026

Abstract

Every collection of fifteen cubes, each displaying all five colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic exterior faces using those five colours. Fourteen need not suffice: a counterexample uses just three varieties, with multiplicities $4, 5, 5$. This determines the five-colour corner value listed as unknown by Berkove, Eickemeyer and Schulman (2018). The lower bound has an elementary proof. For the upper bound, anchored Hall deficiencies and a shared-corner argument reduce the problem to 357 finite cases. A separate checker accepts an interval certificate for every case. Lean proves the interval and packed checkers sound, verifies selected certificates and the fourteen-cube obstruction, and reduces physical counterexamples to the packed root problems. Complete formal verification of the upper bound remains unfinished.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 6 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] lead to a family of extremal questions: how many face-coloured cubes guarantee that eight can be chosen to make a larger cube with monochromatic faces? Each small cube here uses all colours of the specified palette. Colours remain fixed, proper rotations identify varieties, and reflections are not allowed as moves. Collections may contain repeated varieties. Only exterior faces are constrained; touching interior faces need not have equal colours.

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] tabulate the corresponding frame numbers for palettes of two to six colours and leave the five-colour column unknown. Their corner values for two, three, four and six colours are respectively $11, 23, 11, 24$. We determine the missing five-colour value; these thresholds need not increase with the palette size. For a $2 \times 2 \times 2$ cube the frame consists of its eight corners. Let N_5 denote the least integer such that every collection of N_5 five-colour cubes contains a solution using all five colours on the target.

Theorem 1. *The sharp five-colour corner threshold is* $N_5 = 15$.

The conclusion for larger collections follows by choosing any fifteen cubes. The theorem concerns a target that may depend on the collection; it does not assert the ability to construct every target.

The proof has two distinct parts. A short corner argument excludes every target for an explicit fourteen-cube collection. For the upper bound, Hall's theorem gives linear alternatives for failure of a target. We reduce a hypothetical fifteen-cube counterexample to one of 357 cases, then verify finite interval certificates excluding those cases. The certificates, C++ programs, geometric audits, manuscript and Lean sources are included in the accompanying review

archive.[1]

A limited literature search has found no later value for this entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [5] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 2 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the pinned four-to-three-colour project [4].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \dots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \le |D| - 1, \qquad \varnothing \ne D \subseteq \{1, \dots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

[1] Intended repository: https://github.com/jasondavies/macmahon-five-colour-threshold. Public release is pending; the review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 3. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

For an opposite-repeat target, both orientations of each occurring unordered triple must be present. The unordered triples for which both cyclic orientations are available are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies. None is solvable. □

4 A complete cover by 357 cases

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 4 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\text{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m - 1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 4, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If a copies of A and $8 - a$ copies of V build a target, with $a \le m$, then $x_V \le 7 - a$ in a solution-free collection. These

tests are finite: the two varieties must cover all eight target positions, A must fit at least a, and V at least $8 - a$. They are also sufficient: at most a positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to a positions that A fits, and assign the remaining $8 - a$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The reported counts are constraint-orbit counts, not position-subset counts. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection is used only to transport a whole hypothetical counterexample, not as an allowed move on an individual cube. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 5 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the anchor's two two-colour positions cannot be Hall-deficient.*

Proof. Each adjacent-repeat cube has exactly two such types, *aab* and *aac*, with $b \neq c$. There are only $5 \cdot 4 = 20$ types *aab*. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 5 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 6. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

5 Interval certificates

Each case asks whether a vector of nonnegative integers x can satisfy its root bounds, total $G = 15$, root budget and at least one Hall cut for each of the 75 targets. We certify that no such vector exists.

Write a box as $\ell \le x \le u$. For a cut $x(S) \le r$, two necessary conditions for feasibility in the box are

$$\ell(S) \le r, \qquad G \le r + u(S^c). \tag{3}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $s = r - \ell(S)$ and $g = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \le \min(u_i, \ell_i + s) \quad (i \in S), \qquad x_i \ge \max(\ell_i, u_i - g) \quad (i \notin S). \tag{4}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (3). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let s_*, g_* be the largest corresponding s, g. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + s_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - g_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{5}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (4) imply (5). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \ne i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \ne i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (3). These are the checker's 77 possible closing reasons. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \le v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 7 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 357 root cases.*

The archived computation has the following reproducible totals.

Root cases	357
Search tree nodes (splits and leaves)	5,895,461
Certificate instructions (including refinements)	11,599,651
Leaves	2,947,909
Uncompressed certificate bytes	26,203,974
Compressed archive bytes	3,516,042

Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. The witness satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. The fixture is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`. An earlier matching-based exhaustive search also excluded all 357 cases, with 772,590,466 prefix nodes; its rejecting matchings were checked as explicit assignments of eight distinct copies. The interval archive supplies the smaller proof object used here.

Proof of Theorem 1. Lemma 3 gives $N_5 \geq 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 6, and would supply a Hall deficiency for every target. Proposition 7 and the sound interval deductions exclude such a vector. Hence $N_5 \leq 15$. □

6 Formalisation and reproducibility

The package uses Lean 4 and mathlib [6, 7], both at version `v4.35.0`. Lean proves soundness of the reference interval checker, of a faster arithmetic implementation, and of composition of bounded proof chunks. The faster implementation encodes a multiplicity vector as

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad B = 1024.$$

For a neighbourhood S, the weight $W(S) = \sum_{i \in S} B^{74-i}$ satisfies

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Lean proves this identity under $0 \leq x_i \leq 7$, including the absence of carries; the coefficient bound is $75 \cdot 7 < 1024$. It also proves decoding, coordinate updates and equivalence of the packed feasibility test to the integer inequalities. All 19,125 supplied weights are checked against their neighbourhood masks by the Lean kernel.

Chunk composition requires exact agreement of both boundary vectors and a proof excluding each referenced box; the exporter is untrusted. A limit of 128 original instructions gives 109,449 chunks. All 105 certificates with at most 500 instructions have passed kernel replay. Their exact case IDs and instruction counts are in `data/lean-small-replay-cases.json`, tied to the certificate archive by its SHA-256 hash; `make small-replay WORKERS=2` reproduces this check. The full 357-case replay remains in progress. The dependencies of the 59 declarations listed explicitly in `Audit.lean` lie in `propext`, `Classical.choice` and `Quot.sound`; no `native_decide` is used.

Lean proves completeness and uniqueness of the 75 cube varieties and checks all 45,000 compatibility bits using the pinned corner–rotation theorem. The lower bound uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. In the formal proof, this finite check replaces the elementary target classification in Lemma 3; catalogue completeness and rotation transport extend it to every physical target.

Lean proves the physical Hall reduction, symmetry transport, anchor normalisation, two-variety bounds, anchor selection and the shared-corner argument. Finite checks certify all root capacities and cover representatives. These results place every fifteen-cube counterexample in one of the 357 integer root problems, which are proved equivalent to the packed archive cases, including all coordinate bounds, total $G = 15$, root budget and Hall-cut families. In namespace `FiveColour.Packed.Archive`, the exact identifications are `root_model` and `problem_model`, in `FiveColour/ArchiveBridge.lean`.

The checked conditional theorem is `FiveColour.Physical.threshold_of_integer_root_exclusions`, in `FiveColour/CaseProblem.lean`: exclusion of every admissible vector in all 357 root boxes implies the physical fifteen-cube threshold. The physical lower bound is `FiveColour.Physical.lower_bound`, in `FiveColour/LowerBound.lean`. Both appear in the axiom audit. Check these modules and the audit with:

```
lake build FiveColour.CaseProblem FiveColour.LowerBound
lake build FiveColour
python3 -S scripts/audit_lean.py
```

Complete replay and checking the final aggregate remain before Theorem 1 is proved in Lean.

The review archive supplies certificates, code and reproduction commands. A Mathlib-only comparator challenge states the puzzle geometrically. Model equivalence is already proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`; only applying the completed threshold theorem and checking the full comparator solution await replay. Mathematical questions left open here include classifying all fourteen-cube obstructions and determining the five-colour frame numbers for larger cubes.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

References

[1] P. A. MacMahon, *New Mathematical Pastimes*, Cambridge University Press (1921). Digitised edition.

[2] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. doi:10.5281/zenodo.10680216. Journal PDF.

[3] P. Hall, *On Representatives of Subsets*, Journal of the London Mathematical Society **s1-10**(1) (1935), 26–30. doi:10.1112/jlms/s1-10.37.26.

[4] J. Davies, *A sharp threshold for three-colour targets from four-colour cubes*, manuscript and Lean project (2026). Pinned repository revision.

[5] M. Rosenfeld, *MacMahon Cubes and Moshe's Insanity, Puzzle 1*, Geombinatorics Quarterly **36**(2) (October 2026), starts at p. 78. Journal listing; full text not examined.

[6] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[7] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of CPP 2020, ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.